MILITARY SPECIFICATION

ASSEMBLY, PRIORITY LOGIC

65A132F8

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be capable of controlling weapon station piority.

1.2 Classification - Priority logic assembly 65A132F8 shall consist of the item listed in 6.9.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.11 of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfullment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6.4).

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(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including transistors and diodes) shall be obtained as outlined in Specification MIL-E-5400.

3.2.2 Microelectronic Modular Assemblies - When used, microelectronic modular assemblies shall be described by a contractor procurement specification and must be approved by the procuring agency. Microelectronic modular assemblies (microelectronic circuits) (MICS) shall be contained within maintenance modules (for example, on printed circuit cards). The choice of microelectronic modular assemblies shall be based on at least two sources of supply, which can be used interchangeably.

3.2.3 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules (see Specification MIL-S-23603). Maintenance modules shall normally be considered repairable.

3.2.4 Microelectronic Interchangeability - Microelectronic circuits (MICS) shall be interchangeable at the MIC level, or at the maintenance module level when approved by the procuring activity (see 6.13).

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 3.5 ounces.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1,000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 2,000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies,

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and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.4 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3.5 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.6 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.7 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ±1.0 vac and/or 27.5 ±0.5vdc

3.3.8 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for class II equipment, except as modified herein:

3.3.8.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of curve I of Specification MIL-E-5400.

3.3.8.2 Altitude - The unit shall operate satisfactorily when subjected to the altitude requirements of Specification MIL-E-5400 with the exception that the operational altitude be limited to 50,000 feet.

3.3.9 Warm-Up Time - The equipment shall require no warm-up time.

3.3.10 Primary Input Power Requirements - The unit shall operate with the following d-c input power:

(1) DC Power 28V, Category "B", 0.5 amp as specified in MIL-STD-704

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(2) DC Power +10V, 0.5 amp

(3) DC Power -10V, 0.1 amp

3.3.10.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The assembly shall set priorities for release among the various weapons stations on board the aircraft containing stores to be released.

3.4.2 Operational Requirements - The assembly shall meet the operational requirements of 3.4.2.1 through 3.4.2.10, subject to the operating conditions shown on figure 1, and to the following definition of terms:

(1) The logic '0' level shall be less than +0.5 volt dc.

(2) Unless otherwise specified, the logic '1' level shall be greater than +5.0 volts dc.

3.4.2.1 Reset State - With the assembly connected to primary power as shown in figure 1, the closing of switch S16 shall implement the reset state, defined by the following conditions:

(1) The presence on the output at terminal 8, of the logic '0' level for a period of 15 ±10 milliseconds, following which the output shall switch to the logic '1' level. The logic '1' level shall be at its full value, as defined herein, within 30 milliseconds after the reset state is implemented.

(2) The presence, at terminals K and P, of the logic '1' level and at terminals A and 5 of the logic '0' level. These levels shall remain present until the outputs are switched in accordance with 3.4.2.6.1 and 3.4.2.8.1.

3.4.2.1.2 With switch S16 already closed, and with the assembly operating, the reset state shall be alternately implemented by the closing of switch S17. When reset is implemented by this means, the output at terminal 8 shall be at the '0' level for as long as switch S17 is closed, and shall return to '1' level when the switch is subsequently opened.

3.4.2.2 Terminal 6. The output at terminal 6 shall be at the logic '0' level when switch S18 is closed, and shall otherwise be at the logic '1' level.

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3.4.2.3 Terminal H. The output at terminal H shall be at the logic '1' level when switch S13 is closed and shall otherwise be at the logic '0' level.

3.4.2.4 Terminal N. The output at terminal N shall be at the logic '1' level when switch S14 is closed, and shall otherwise be at the logic '0' level.

3.4.2.5 Terminal T. The output at terminal T shall be at the logic '1' level when switch S15 is closed and switch S2 is open, and shall otherwise be at the '0' level. Switch-over from the '0' to the '1' logic level shall occur between 0.1 and 0.3 millisecond after the last required input condition is implemented.

3.4.2.6 Terminal 5.

3.4.2.6.1 '0' to '1'. The output at terminal 5 shall switch from the logic '0' level when switch S15 is closed and the following conditions are already present:

(1) Switches S2 and S4 are open.

(2) Switch S14 is closed.

(3) The output at terminal K is at the logic '1' level. Switching shall occur $3^{+2.0}_{-0.5}$ milliseconds after the necessary conditions are fulfilled.

3.4.2.6.2 '1' to '0'. The output at terminal 5 shall return to the logic '0' level when the reset state is implemented, or when switch S5, S6, S7 or S8 is opened with the other three switches already open.

3.4.2.7 Terminal K. The output at terminal K shall be the logic inverse of the output at terminal 5.

3.4.2.8 Terminal A.

3.4.2.8.1 '0' to '1'. The output at terminal A shall switch from the logic '0' to the logic '1' level when switch S15 is closed and the following conditions are already present:

(1) Switches S2 and S3 are open.

(2) Switch S14 is closed.

(3) The output at terminal K is at the logic '1' level. Switching shall occur 2.0 ±1 milliseconds after the necessary conditions are fulfilled.

3.4.2.8.2 '1' to '0'. The output at terminal A shall return to the logic '0' level when any of the following conditions or sets of conditions is satisfied:

(1) The reset state is implemented.

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(2) Switch 9, 10, 11, or 12 is opened with the others already open; or

(3) One or more of the following conditions occurs, with the others already in existence:

(a) Switch S3 is closed.

(b) Switch S4 is opened; and

(c) Terminal K output is at the logic '1' level.

3.4.2.9 Terminal P. The output at terminal P shall be the logic inverse of the output at terminal A.

3.4.2.10 Terminal 19. The output at terminal 19 shall be of the logic '0' level when the following two conditions are ANDed: (1) Switch S2 is open; and (2) Switch S3 or S4 (or both) are open or Switch S1 is closed. At all other times, the output shall be at the logic '1' level.

3.5 Detail Requirements - Priority logic assembly 65A132F8 shall meet the following requirements:

3.5.1 Function - The assembly sets priorities for release among the various weapons stations on board the aircraft containing stores to be released.

3.5.2 Form Factor - The priority logic assembly shall conform to the dimensions of figure 2.

3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see 6.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction tests

(2) Initial production tests

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(3) Acceptance tests

(4) Life tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

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4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual tests

(2) Sampling tests

(3) Reliability assurance tests

(4) Special tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of product

(2) Operational test

(3) Manufacturing run in test

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4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless reliability assurance tests of 4.4.3 are deleted by contractual action.)

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Quantity of Equipments Offered for Acceptance	Quantity to Be Selected for Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the initial production test (4.3) is deleted, then a sample test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause a remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by MIL-STD-781. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Test Level - The Test Level shall be F from Table I of MIL-STD-781.

4.4.3.2 Accent-Reject Criteria - Test plan V of MIL-STD-781 shall be used to determine the accept-reject criteria for the reliability qualification test plan. Test plan VI of MIL-STD-781 shall be used to determine the accept-reject criteria for the reliability sampling test plan.

4.4.3.3 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 2 hours during each cycle.

4.4.3.4 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.10.

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4.4.3.5 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance charactertics fall below the acceptance requirement (4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.6 Preventive Maintenance - During the period of the tests no preventive maintenance measures are required.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be

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performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the initial production test, sampling tests, reliability tests, or special tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to Be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft)
Humidity	Room ambient
AC voltage	115 ±5 volts (at lowest applicable frequency)
DC voltage	27.0 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) Circuit reference symbol number

(3) Part function

(4) Name of manufacturer

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(5) Nature of failure

(6) Number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - Equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The equipment shall be used in conjunction with the items listed in 6.9.1 to fulfill the requirements of Programmer, Aircraft Weapons Release Set MX-7016(*)/AWE-1 as described in Specification MIL-P-81361(AS).

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 vdc and 115 ±1.0 vac as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

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(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.6. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.6).

6.8 Asterisk - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.9 Items of Equipment - The equipment covered by this specification shall consist of the following item:

Item	Type Designation	Appl. Para.
Assembly, Priority Logic	65A132F8	3.5

6.9.1 Associated Equipment - The equipment shall operate with the following associated equipment:

(1) Assembly, Left-Right Station 65A132F5

(2) Assembly, Center Station 65A132F6

(3) Assembly, Station Transfer 65A132F7

(4) Assembly, Quantity Select 65A132F9

6.10 Performance Characteristics to Be Measured - The equipment shall meet the performance characteristics specified in 6.10.7.1 through 6.10.7.13.

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6.10.1 Methods of Test - The exact test method is dependent upon the equipment available. Equivalent methods of test are acceptable provided they meet the accuracy and performance requirements of this specification. It shall also be permissible to combine tests where operational capabilities of test equipment allow it.

6.10.2 Standard Switch Positions - The positions shown in figure 1 are the normally OFF positions of all the switches. The switches shall have a maximum contact bounce of 1 millisecond with the exception of S1, S2, S3 and S4, which should have zero bounce.

6.10.3 Standard Load - The standard load shall be as specified in figure 1.

6.10.4 Standard Connections - The standard loads, signal sources and power supplies shall be connected as shown in figure 1.

6.10.5 Test Procedure

(1) Before actual test procedures are started, a card shall be examined for shorts of less than 10 ohms between all connected pins with the following exceptions: Pins 20 to X, 21 to Y, and 22 to Z.

(2) Where reset is indicated, switch S17 shall be closed and immediately returned to its normally open position.

(3) All switches shall be in their open positions unless they are called out to be closed.

(4) After each test is performed, all inputs shall be removed prior to commencing of next test, unless it is a sequential type test.

6.10.6 Logic-Level Reference - A voltage specified as a logical "0" shall not exceed +0.5 vdc, and a voltage specified as a logical "1" shall be not less than +5 vdc nor more than the positive supply voltage.

6.10.7 Individual Tests - The following tests shall be performed on each assembly:

6.10.7.1 Diode Leakage Test - With +28 vdc applied to pins 1, 2, 3, 4, B, C, D, E, N, 8 and S and pin Z grounded, the current through each pin shall not exceed 50 ua.

6.10.7.2 Diode Short Circuit Test - With switches S1, S16, S3 and S4 closed, the current through switches S3 and S4 shall not exceed 3 ma, respectively.

6.10.7.3 Reset Test

(1) Connect equipment as shown in figure 3.

(2) Trigger oscilloscope externally with + slope.

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(3) Close switch S16.

(4) Observe "O" for first 15 ±10 ms.

(5) Observe "1" at 30 ms.

(6) Switch S16 is to remain closed for all subsequent tests.

6.10.7.4 Salvo Test

(1) Observe "1" on pin 6.

(2) Close switch S18.

(3) Observe "O" on pin 6.

6.10.7.5 Pairs Test

(1) Observe "O" on pin H.

(2) Close switch S13.

(3) Observe "1" on pin H.

6.10.7.6 Singles Test

(1) Observe "O" on pin N.

(2) Close switch S14.

(3) Observe "1" on pin N.

6.10.7.7 Tone Test

(1) Close switches S1, S3, and S4.

(2) Observe "O" on pin 19.

(3) Open switch S1.

(4) Observe pin 19 > 20 vdc.

(5) Open switch S3.

(6) Observe "O" on pin 19.

(7) Close switch S3.

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(8) Observe pin 19 > 20 vdc.

(9) Open switch S4.

(10) Observe "O" on pin 19.

(11) Close switch S4.

(12) Observe pin 19 > 20 vdc.

(13) Close switch S2 and close switch S1.

(14) Observe pin 19 > 20 vdc.

6.10.7.8 Memory Test

(1) Reset.

(2) Observe "O" on pin 5.

(3) Observe "1" on pin K.

(4) Observe "O" on pin A.

(5) Observe "1" on pin P.

(6) Close switch S14.

(7) Close switch S15.

(8) Observe "1" on pin 5.

(9) Observe "O" on pin K.

(10) Observe "1" on pin A.

(11) Observe "O" on pin P.

6.10.7.9 Left Memory Test

(1) Reset.

(2) Close switch S14 and switch S3

(3) Close switch S15.

(4) Observe "1" on pin 5.

(5) Observe "O" on pin K.

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(6) Observe "O" on pin A.

(7) Observe "1" on pin P.

(8) Open switch S15.

(9) Close switch S5.

(10) Open switch S5.

(11) Observe "O" on pin 5.

(12) Observe "1" on pin K.

(13) Close switch S15.

(14) Open switch S15.

(15) Close switch S6.

(16) Open switch S6.

(17) Observe "O" on pin 5.

(18) Observe "1" on pin K.

(19) Close switch S15.

(20) Open switch S15.

(21) Close switch S7.

(22) Open switch S7.

(23) Observe "O" on pin 5.

(24) Observe "1" on pin K.

(25) Close switch S15.

(26) Open switch S15.

(27) Close switch S8.

(28) Open switch S8.

(29) Observe "O" on pin 5.

(30) Observe "1" on pin K.

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6.10.7.10 Right Memory Test

(1) Reset.

(2) Close switch S14 and switch S4.

(3) Close switch S15.

(4) Observe "O" on pin 5.

(5) Observe "1" on pin K.

(6) Observe "1" on pin A.

(7) Observe "O" on pin P.

(8) Open switch S15.

(9) Close switch S9.

(10) Open switch S9.

(11) Observe "O" on pin A.

(12) Observe "1" on pin P.

(13) Close switch S15.

(14) Open switch S15.

(15) Close switch S10.

(16) Open switch S10.

(17) Observe "O" on pin A.

(18) Observe "1" on pin P.

(19) Close switch S15.

(20) Open switch S15.

(21) Close switch S11.

(22) Open switch S11.

(23) Observe "O" on A.

(24) Observe "1" on P.

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(25) Close switch S15.

(26) Open switch S15.

(27) Close switch S12.

(28) Open switch S12.

(29) Observe "O" on A.

(30) Observe "1" on P.

6.10.7.11 Right Memory Bypass Test

(1) Connect equipment as shown in figure 4.

(2) Trigger oscilloscope externally with a - slope.

(3) Reset.

(4) Close switch S14 and switch S4.

(5) Close switch S15.

(6) Open switch S15 and switch S4.

(7) Observe "O" on pin P.

(8) Observe "1" on pin A.

(9) Close switch S3.

(10) Observe "O" for first 2.0 $\pm$1 ms on pin P.

(11) Observe "1" at 5 ms on pin P.

6.10.7.12 Transfer Test

(1) Connect equipment as shown in figure 5.

(2) Trigger oscilloscope externally on + slope.

(3) Reset.

(4) Close switch S2 and switch S14.

(5) Close switch S15.

(6) Observe "O" on pin 5.

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(7) Observe "1" on pin K.

(8) Observe "O" on pin A.

(9) Observe "1" on pin P.

(10) Open switch S15.

(11) Open switch S2.

(12) Close switch S15.

(13) Observe "O" on pin 5 for 3 ms $^{+2.0}_{-0.5}$ ms.

(14) Observe "1" on pin 5 at 10 ms.

6.10.7.13 Fire Pulse Test

(1) Connect equipment as shown in figure 6.

(2) Trigger oscilloscope externally with + slope.

(3) Reset.

(4) Close switch S2 and switch S14.

(5) Close switch S15.

(6) Observe "O" on pin T.

(7) Open switch S15.

(8) Open switch S2.

(9) Close switch S15.

(10) Observe "O" on pin T for first 200 $\pm$100 μs.

(11) Observe "1" on pin T at 400 μs.

6.11 Applicable Documents -

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
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Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-23603	System Readiness/Maintainability; Avionic Systems Design, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution

PUBLICATION

Naval Air Systems Command

MIL-P-81361(AS)	Programmer, Aircraft Weapons Release Set MX-7016(*)/AWE-1

6.11.1 Availability of Documents -

(1) When requested specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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6.12 Criteria for Multiple Source - The criteria of this paragraph shall be used for determining the extent to which the requirement for multiple independent source availability of microcircuits is satisfied. For interchangeability at the MIC level, multiple source availability exists if the devices from the different sources may be used interchangeably, device by device, in the final equipment without any need for mechanical or electrical changes. For interchangeability at the card level, multiple source availability exists if a card composed of MICS from one source is mechanically and electrically interchangeable with another card composed of MICS from another source. Numerous aspects need be considered to arrive at the independence of sources; these involve production independence, corporate (or firm) and financial independence, and geographical separation. The production independence of one source of another is determined by the extent to which the one source is independent of the other for manufacture; and by the lack of common original suppliers of raw or processed materials. Design, process information, technology, and diffusion masks may be derived from a common source. However, in such a case, each device source shall have a demonstrated capability of originating these items. The corporate (or firm) and financial independence of one source of another is determined by the absence of common directors, common officers, or holders of large common financial interests. Geographical separation is simply the distance between the locations where principal manufacture takes place by two sources.

6.13 Microelectronic Interchangeability Demonstration - Interchangeability of microelectronic elements shall be demonstrated as follows:

(1) For the interchangeability at the MIC level, it shall be demonstrated for each type of functional MIC from one source, as used in the system, that it is mechanically and electrically interchangeable with that from another source (see 6.12).

(2) For the interchangeability at the maintenance module level, it shall be demonstrated that modules typical of those to be used in the system composed of MICS from one source are mechanically and electrically interchangeable with modules composed of MICS from another source.

(3) For the completed equipment, it shall be demonstrated that modules composed of MICS from one source are interchangeable with corresponding modules composed of MICS from another source. When interchange has been effected, no degradation shall take place in system and subsystem performance.

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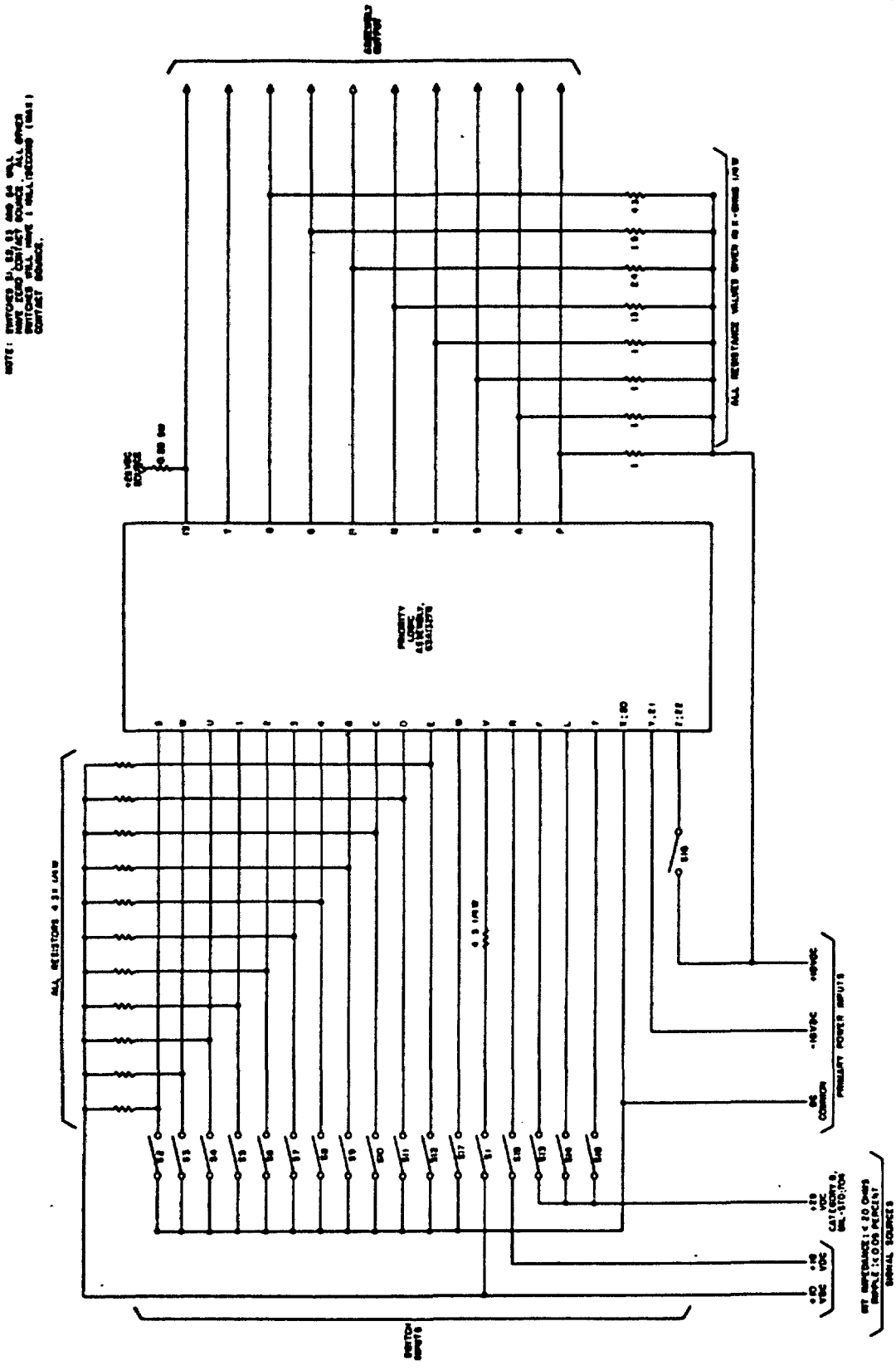


Figure 1. Operational Test Setup

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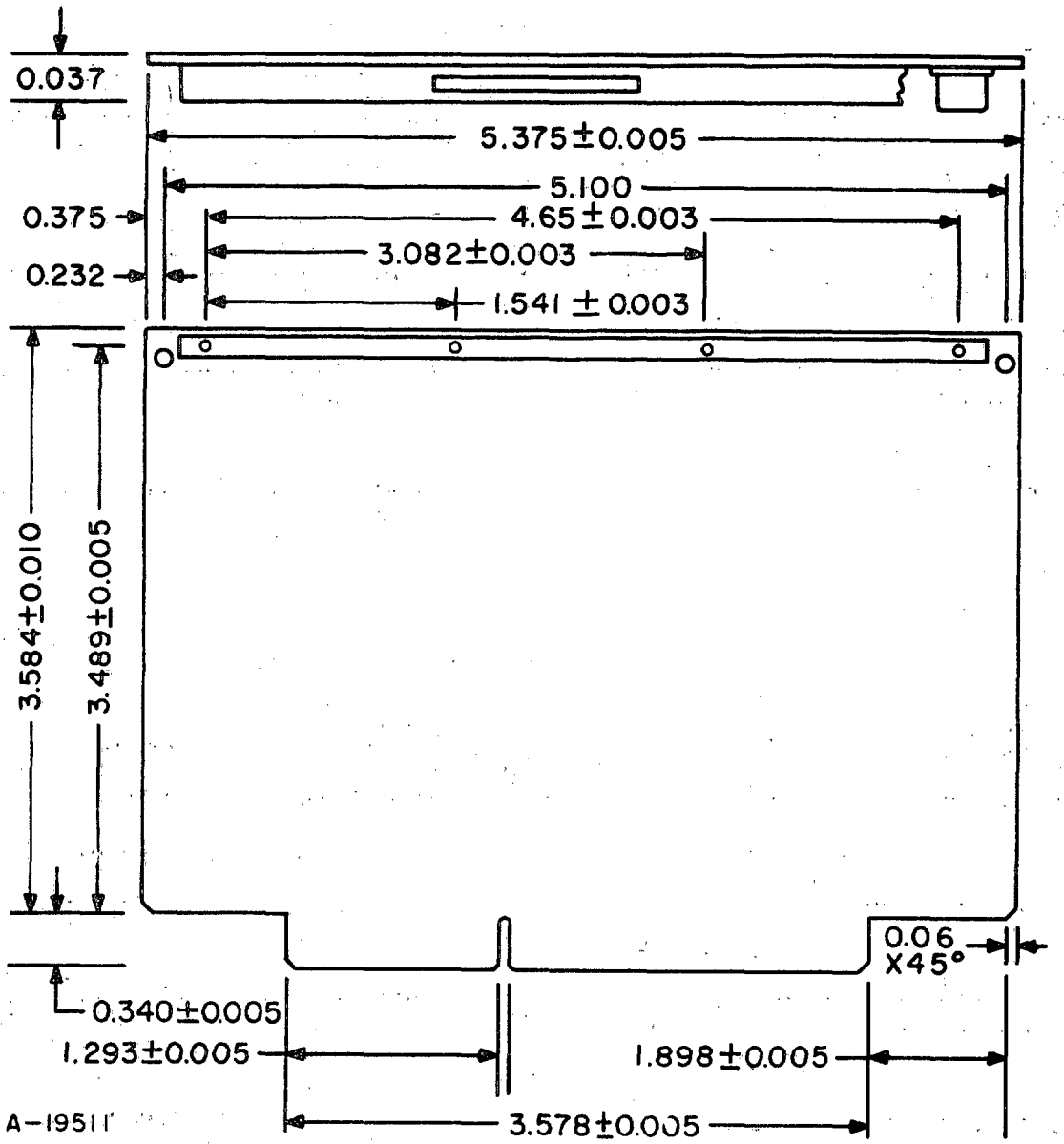


Figure 2. Outline Dimensions

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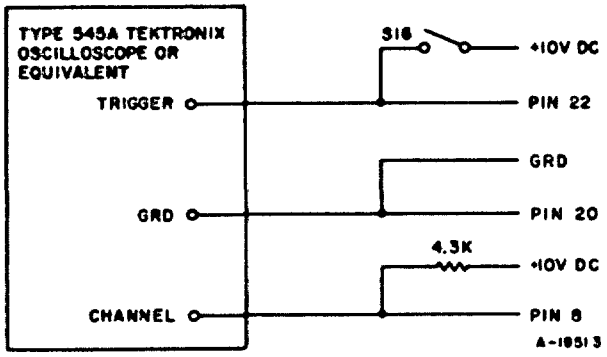


Figure 3. Test Arrangement 1

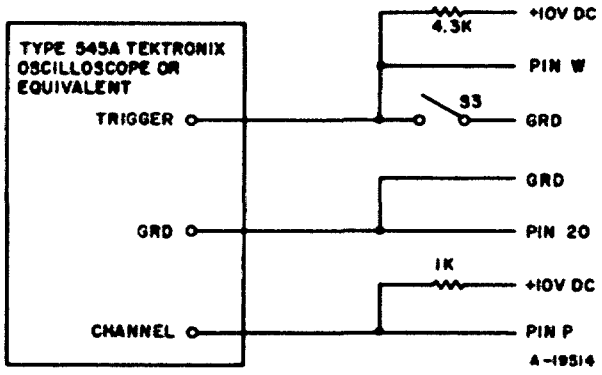


Figure 4. Test Arrangement 2

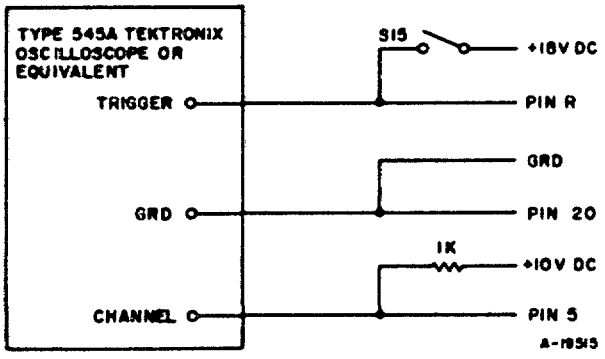


Figure 5. Test Arrangement 3

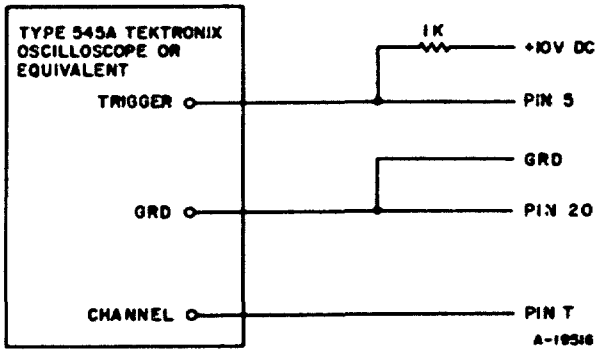


Figure 6. Test Arrangement 4

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-A-81372(AS) ASSEMBLY, PRIORITY LOGIC 65A132F8

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

FORM 1426
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Naval Air Systems Command
Washington, D.C. 20360

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